Do Now · retrieval

Lesson 141 · ACID Transactions, Locking and Redundancy

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does foreign-key referential integrity prevent?

Q2

Which is faster in the worst case on sorted arrays, linear or binary search?

Q3

Which logic gate outputs 1 only when both inputs are 1?

Q4

What is the accumulator for?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Begin transaction — Move £20: subtract account A and add account B.

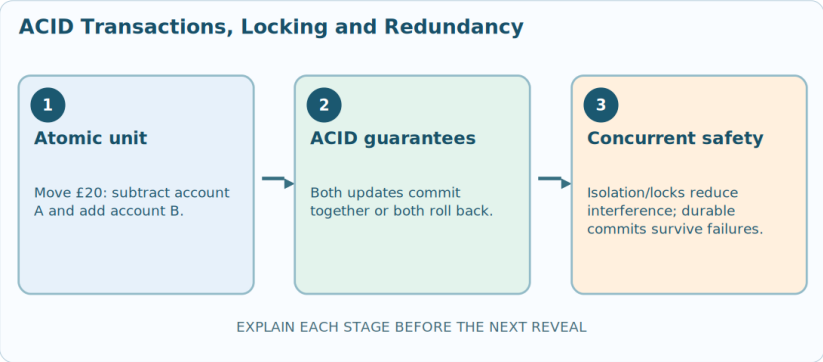
Core idea 2

Protect consistency — Both updates commit together or both roll back.

Core idea 3

Coordinate concurrent work — Isolation/locks reduce interference; durable commits survive failures.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

transaction — Group of DB operations treated as one unit of work.

ACID — Atomicity, Consistency, Isolation, Durability.

record locking — Restricting conflicting access while data is being changed.

redundancy — Duplicated resources/data for resilience, distinct from backup.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

A transfer subtracts £20 from Account A but crashes before adding £20 to Account B. Which ACID property matters most?

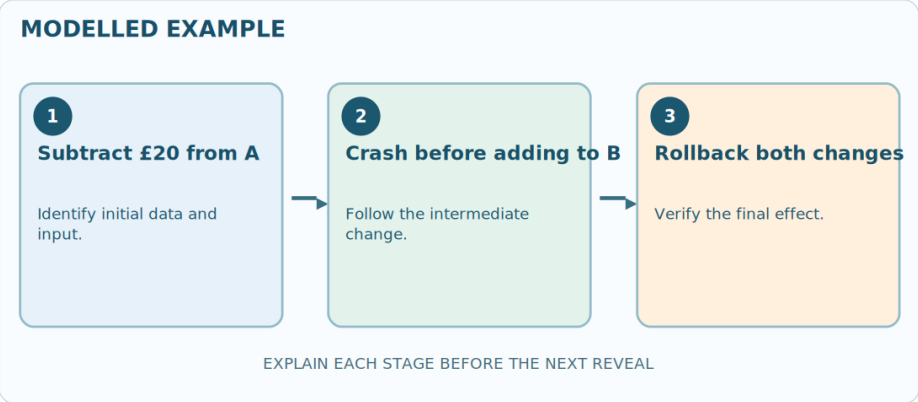
Follow the worked method

1. Subtract £20 from A — Identify the relevant information and make a prediction.

2. Crash before adding to B — Explain the change using correct technical terms.

3. Rollback both changes — Atomicity means the transfer is all-or-nothing. A DBMS transaction can roll back partial changes. Consistency checks valid constraints; isolation keeps concurrent transfers safe; durability preserves committed transfers.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Two concurrent requests
Identify the relevant information and make a prediction.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Expand ACID and explain atomicity and durability. [6]

Question 2

Explain why two concurrent users editing one record can cause a lost update. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate replication, RAID and backups for database resilience. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

RAID or replication guarantees recovery after an accidental DELETE.

Use key vocabulary from memory

1. Explain transaction in your own words.

2. Explain ACID in your own words.

3. Explain record locking in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.